2026

NETWORK512

FORMATIVE ASSESSMENT 3

MOAGI MMOLAWA
20261359

WEEK 1: The IOS Operating System

Topology



User EXEC mode

Terminal

```
Switch> ?
Exec commands:
  connect     Open a terminal connection
  disable     Turn off privileged commands
  disconnect  Disconnect an existing network connection
  enable      Turn on privileged commands
  exit        Exit from the EXEC
  logout      Exit from the EXEC
  ping        Send echo messages
  resume      Resume an active network connection
  show        Show running system information
  ssh         Open a secure shell client connection
  telnet      Open a telnet connection
  terminal    Set terminal line parameters
  traceroute  Trace route to destination
Switch> show ?
  arp                 Arp table
  cdp                 CDP information
  clock               Display the system clock
  crypto              Encryption module
  dtp                 DTP information
  etherchannel        EtherChannel information
  flash:              display information about flash: file system
  history             Display the session command history
  interfaces          Interface status and configuration
  ip                  IP information
  lldp                LLDP information
  mac                 MAC configuration
  mac-address-table   MAC forwarding table
  mls                 Show MultiLayer Switching information
  privilege           Show current privilege level
  sessions            Information about Telnet connections
  ssh                 Status of SSH server connections
  tcp                 Status of TCP connections
  terminal            Display terminal configuration parameters
  users               Display information about terminal lines
  version             System hardware and software status
  vlan                VTP VLAN status
  vtp                 VTP information
Switch> show version
Cisco IOS Software, C2960 Software (C2960-LANBASEK9-M), Version 15.0(2)SE4, RELEASE SOFTWARE (fc1)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2013 by Cisco Systems, Inc.
Compiled Wed 26-Jun-13 02:49 by mnguyen

ROM: Bootstrap program is C2960 boot loader
```

Priviledged EXEC and Global Configuration

```
Switch> enable
Switch# configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)# ?
Configure commands:
  aaa                 Authentication, Authorization and Accounting.
  access-list         Add an access list entry
  banner              Define a login banner
  boot                Boot Commands
  cdp                 Global CDP configuration subcommands
  clock               Configure time-of-day clock
  crypto              Encryption module
  default             Set a command to its defaults
  do-exec             To run exec commands in config mode
  dot1x               IEEE 802.1X Global Configuration Commands
  enable              Modify enable password parameters
  end                 Exit from configure mode
  exit                Exit from configure mode
  hostname            Set system's network name
  interface           Select an interface to configure
  ip                  Global IP configuration subcommands
  line                Configure a terminal line
  lldp                Global LLDP configuration subcommands
  logging             Modify message logging facilities
  mac                 MAC configuration
  mls                 mls global commands
 --More--
```

Configuration Management

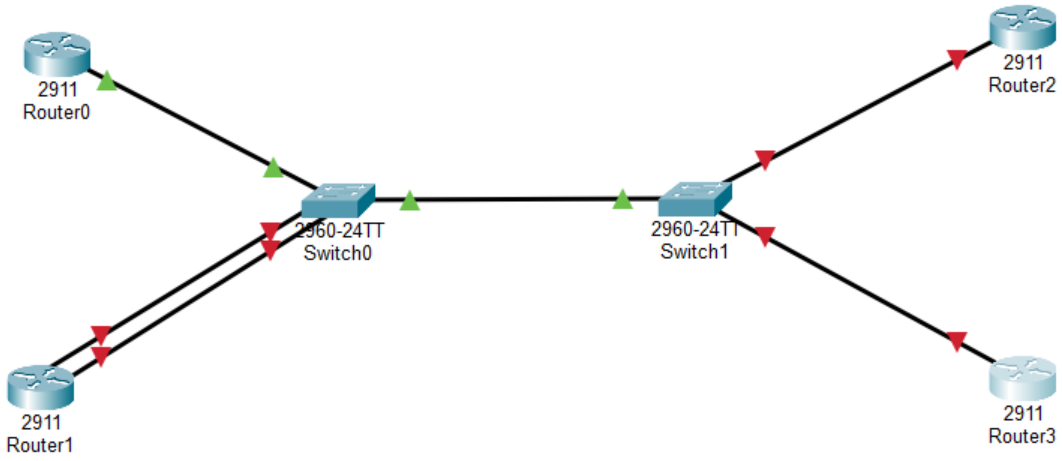
```
Switch(config)# end
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch# show running-config
Building configuration...

Current configuration : 1080 bytes
!
version 15.0
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname Switch
!
!
!
!
!
!
spanning-tree mode pvst
spanning-tree extend system-id
!
interface FastEthernet0/1
!
interface FastEthernet0/2

Switch# copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
Switch# show startup-config
Using 1080 bytes
!
version 15.0
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname Switch
!
```

WEEK 2: Cisco Device Functions



Task 5

Router0

Physical Config CLI Attributes

IOS Command Line Interface

```
%LINK-3-UPDOWN: Interface Vlan1, changed state to down

%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

%LINK-3-UPDOWN: Interface GigabitEthernet0/1, changed state to down

%LINK-3-UPDOWN: Interface GigabitEthernet0/2, changed state to down

%LINK-5-CHANGED: Interface Vlan1, changed state to administratively down

%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to administratively down

%LINK-5-CHANGED: Interface GigabitEthernet0/2, changed state to administratively down

%SYS-5-CONFIG_I: Configured from console by console


0>enable
Password:
Password:
Password:
% Bad secrets

0> enable
Password:
Password:
Password:
% Bad secrets

0> show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set
```

Router1

Physical Config CLI Attributes

IOS Command Line Interface

```
third-party authority to import, export, distribute or use encryption.
Importers, exporters, distributors and users are responsible for
compliance with U.S. and local country laws. By using this product you
agree to comply with applicable laws and regulations. If you are unable
to comply with U.S. and local laws, return this product immediately.

A summary of U.S. laws governing Cisco cryptographic products may be found at:
http://www.cisco.com/wwl/export/crypto/tool/stqrg.html

If you require further assistance please contact us by sending email to
export@cisco.com.

Cisco CISCO2911/K9 (revision 1.0) with 491520K/32768K bytes of memory.
Processor board ID FTX152400KS
3 Gigabit Ethernet interfaces
DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)


         --- System Configuration Dialog ---

Would you like to enter the initial configuration dialog? [yes/no]: show ip route
% Please answer 'yes' or 'no'.
Would you like to enter the initial configuration dialog? [yes/no]: no


Press RETURN to get started!



Router>enable
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set
```

Router2

Physical Config CLI Attributes

IOS Command Line Interface

```
States and local country laws governing import, export, transfer and
use. Delivery of Cisco cryptographic products does not imply
third-party authority to import, export, distribute or use encryption.
Importers, exporters, distributors and users are responsible for
compliance with U.S. and local country laws. By using this product you
agree to comply with applicable laws and regulations. If you are unable
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Processor board ID FTX152400KS
3 Gigabit Ethernet interfaces
DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)


         --- System Configuration Dialog ---

Would you like to enter the initial configuration dialog? [yes/no]: no


Press RETURN to get started!



Router>enable
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set
```

Router3

Physical Config CLI Attributes

IOS Command Line Interface

```
States and local country laws governing import, export, transfer and
use. Delivery of Cisco cryptographic products does not imply
third-party authority to import, export, distribute or use encryption.
Importers, exporters, distributors and users are responsible for
compliance with U.S. and local country laws. By using this product you
agree to comply with applicable laws and regulations. If you are unable
to comply with U.S. and local laws, return this product immediately.

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http://www.cisco.com/wwl/export/crypto/tool/stqrg.html

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Processor board ID FTX152400KS
3 Gigabit Ethernet interfaces
DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)


         --- System Configuration Dialog ---

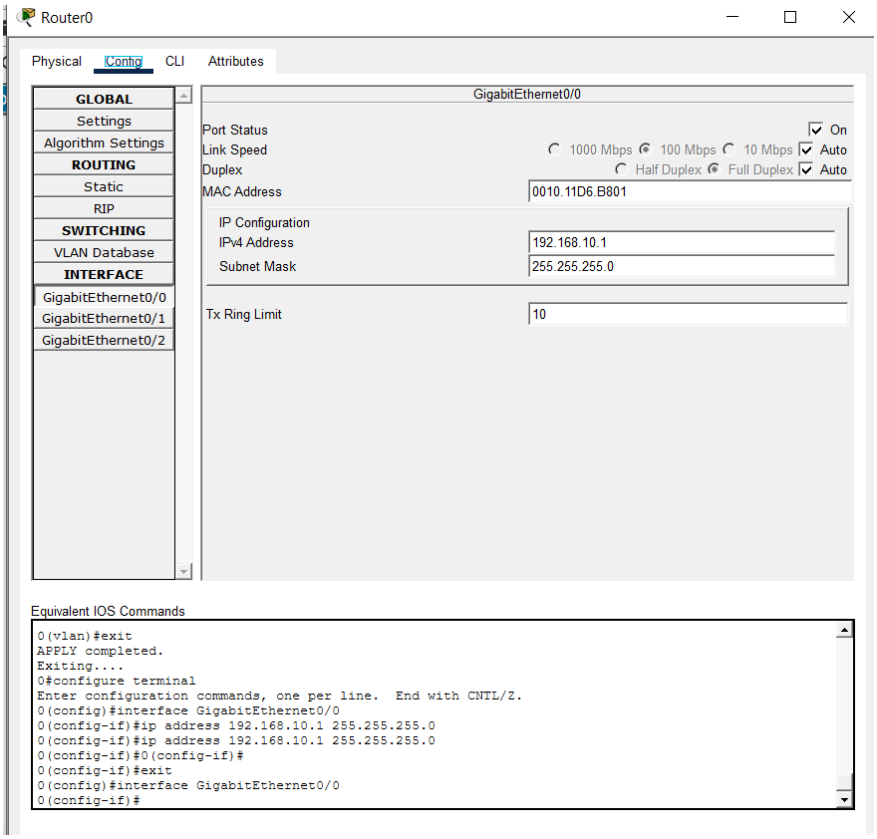
Would you like to enter the initial configuration dialog? [yes/no]: no

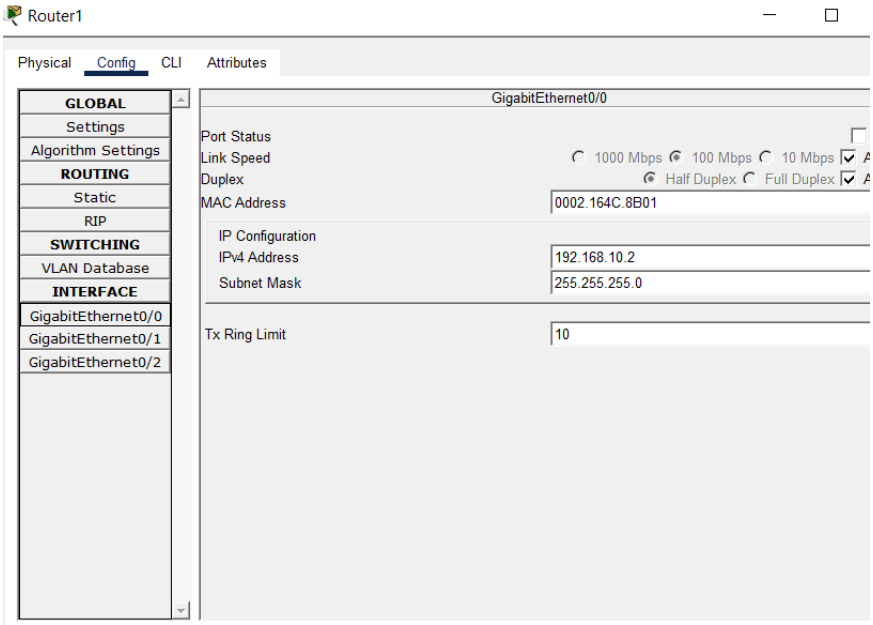

Press RETURN to get started!


Router>enable
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set
```

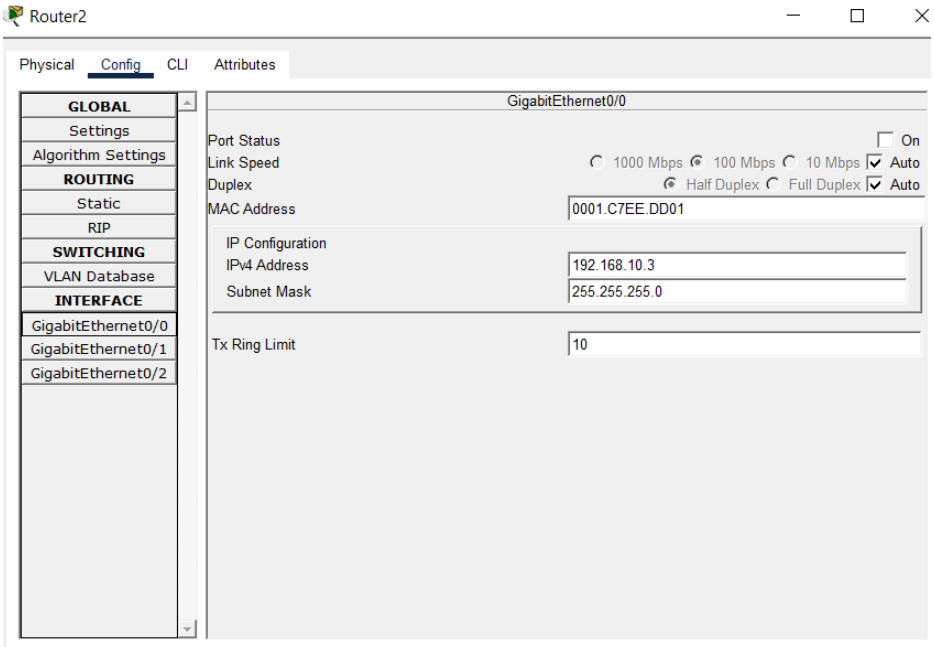
Task 6





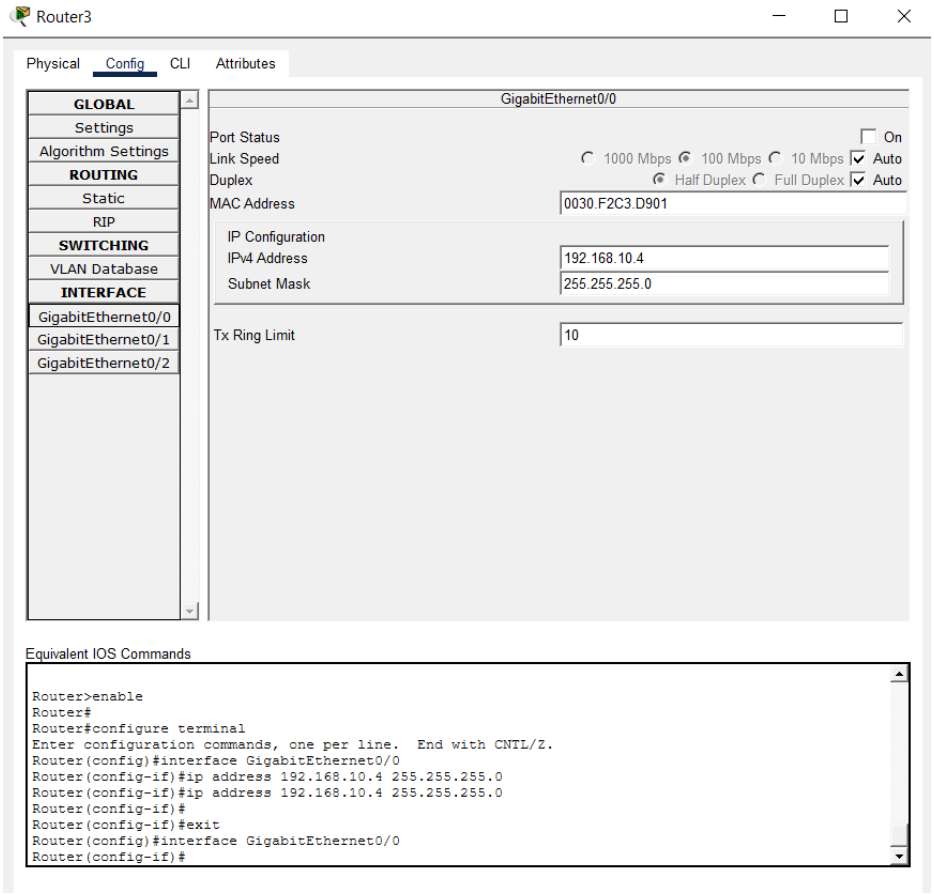
Equivalent IOS Commands

```
Router>enable
Router#
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface GigabitEthernet0/0
Router(config-if)#ip address 192.168.10.2 255.255.255.0
Router(config-if)#ip address 192.168.10.2 255.255.255.0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/0
Router(config-if)#
```

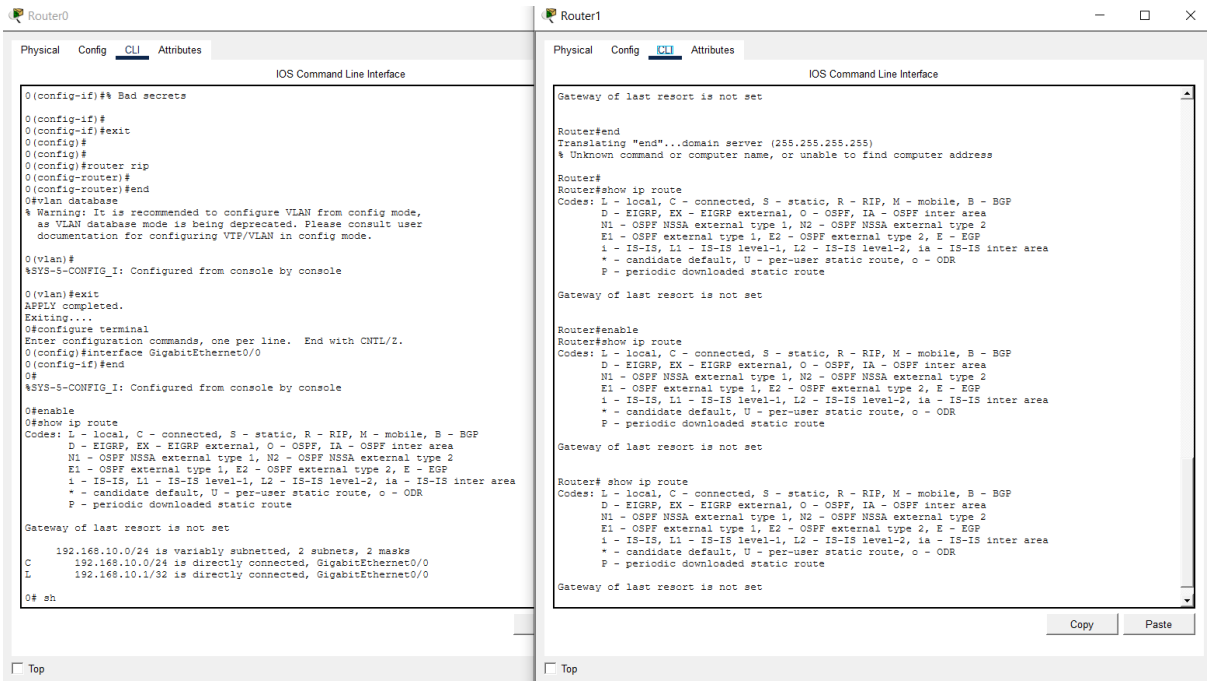


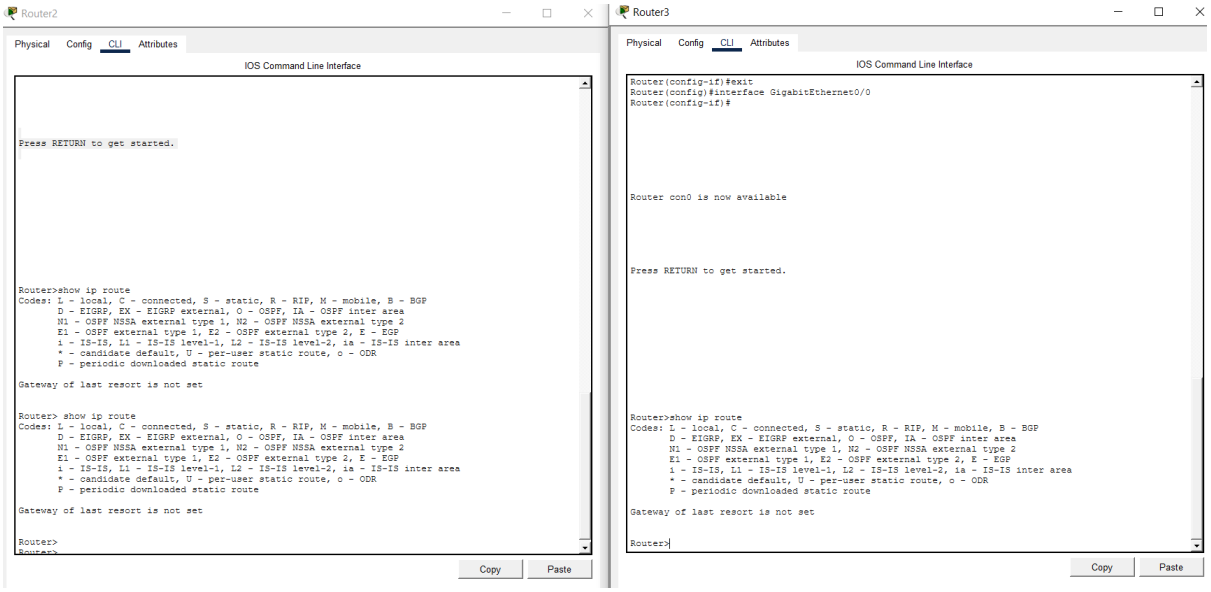
Equivalent IOS Commands

```
Router>enable
Router#
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface GigabitEthernet0/0
Router(config-if)#ip address 192.168.10.3 255.255.255.0
Router(config-if)#ip address 192.168.10.3 255.255.255.0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/0
Router(config-if)#
```

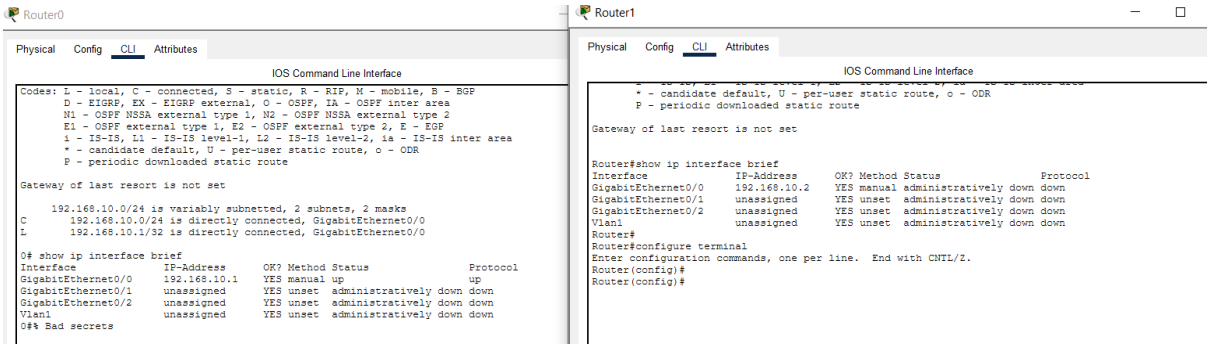

Router3
Physical
Config
CLI
Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
SWITCHING
VLAN Database
INTERFACE
GigabitEthernet0/0
GigabitEthernet0/1
GigabitEthernet0/2
GigabitEthernet0/0
Port Status
On
Link Speed
1000 Mbps
100 Mbps
10 Mbps
Auto
Duplex
Half Duplex
Full Duplex
Auto
MAC Address
0030.F2C3.D901
IP Configuration
IPv4 Address
192.168.10.4
Subnet Mask
255.255.255.0
Tx Ring Limit
10
Equivalent IOS Commands
Router>enable
Router#
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface GigabitEthernet0/0
Router(config-if)#ip address 192.168.10.4 255.255.255.0
Router(config-if)#ip address 192.168.10.4 255.255.255.0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/0
Router(config-if)#

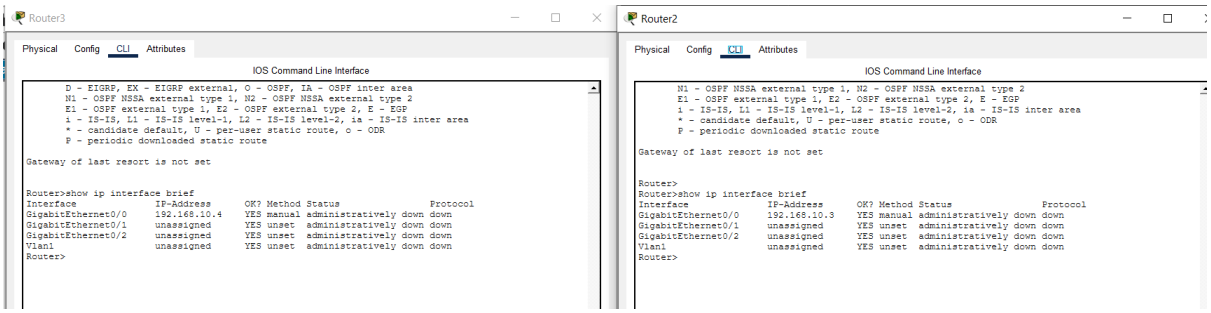
Task 7


Router0
Physical
Config
CLI
Attributes
IOS Command Line Interface
0(config-if)#% Bad secrets
0(config-if)#
0(config-if)#exit
0(config)#
0(config)#
0(config)#router rip
0(config-router)#
0(config-router)#end
0#vlan database
% Warning: It is recommended to configure VLAN from config mode,
as VLAN database mode is being deprecated. Please consult user
documentation for configuring VTP/VLAN in config mode.
0(vlan)#
%SYS-5-CONFIG_I: Configured from console by console
0(vlan)#exit
APPLY completed.
Exiting....
0#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
0(config)#interface GigabitEthernet0/0
0(config-if)#end
0#
%SYS-5-CONFIG_I: Configured from console by console
0#enable
0#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route
Gateway of last resort is not set
192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C 192.168.10.0/24 is directly connected, GigabitEthernet0/0
L 192.168.10.1/32 is directly connected, GigabitEthernet0/0
0# sh
Top
Router1
Physical
Config
CLI
Attributes
IOS Command Line Interface
Gateway of last resort is not set
Router#end
Translating "end"...domain server (255.255.255.255)
% Unknown command or computer name, or unable to find computer address
Router#
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route
Gateway of last resort is not set
Router#enable
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route
Gateway of last resort is not set
Router# show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
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* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route
Gateway of last resort is not set
Copy
Paste
Top


Router2
Physical Config CLI Attributes
IOS Command Line Interface
Press RETURN to get started.
Router>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route
Gateway of last resort is not set
Router> show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route
Gateway of last resort is not set
Router>
Router>
Copy
Paste
Router3
Physical Config CLI Attributes
IOS Command Line Interface
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/0
Router(config-if)#
Router con0 is now available
Press RETURN to get started.
Router>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route
Gateway of last resort is not set
Router>
Copy
Paste

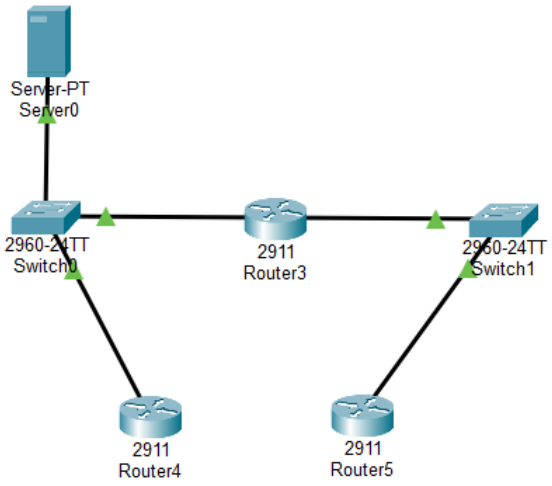
Task 8


Router0
Physical Config CLI Attributes
IOS Command Line Interface
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route
Gateway of last resort is not set
192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C 192.168.10.0/24 is directly connected, GigabitEthernet0/0
L 192.168.10.1/32 is directly connected, GigabitEthernet0/0
0# show ip interface brief
Interface IP-Address OK? Method Status Protocol
GigabitEthernet0/0 192.168.10.1 YES manual up up
GigabitEthernet0/1 unassigned YES unset administratively down down
GigabitEthernet0/2 unassigned YES unset administratively down down
Vlan1 unassigned YES unset administratively down down
0#% Bad secrets
Router1
Physical Config CLI Attributes
IOS Command Line Interface
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route
Gateway of last resort is not set
Router#show ip interface brief
Interface IP-Address OK? Method Status Protocol
GigabitEthernet0/0 192.168.10.2 YES manual administratively down down
GigabitEthernet0/1 unassigned YES unset administratively down down
GigabitEthernet0/2 unassigned YES unset administratively down down
Vlan1 unassigned YES unset administratively down down
Router#
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#
Router(config)#


Router3
Physical Config CLI Attributes
IOS Command Line Interface
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route
Gateway of last resort is not set
Router>show ip interface brief
Interface IP-Address OK? Method Status Protocol
GigabitEthernet0/0 192.168.10.4 YES manual administratively down down
GigabitEthernet0/1 unassigned YES unset administratively down down
GigabitEthernet0/2 unassigned YES unset administratively down down
Vlan1 unassigned YES unset administratively down down
Router>
Router2
Physical Config CLI Attributes
IOS Command Line Interface
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route
Gateway of last resort is not set
Router>
Router>show ip interface brief
Interface IP-Address OK? Method Status Protocol
GigabitEthernet0/0 192.168.10.3 YES manual administratively down down
GigabitEthernet0/1 unassigned YES unset administratively down down
GigabitEthernet0/2 unassigned YES unset administratively down down
Vlan1 unassigned YES unset administratively down down
Router>

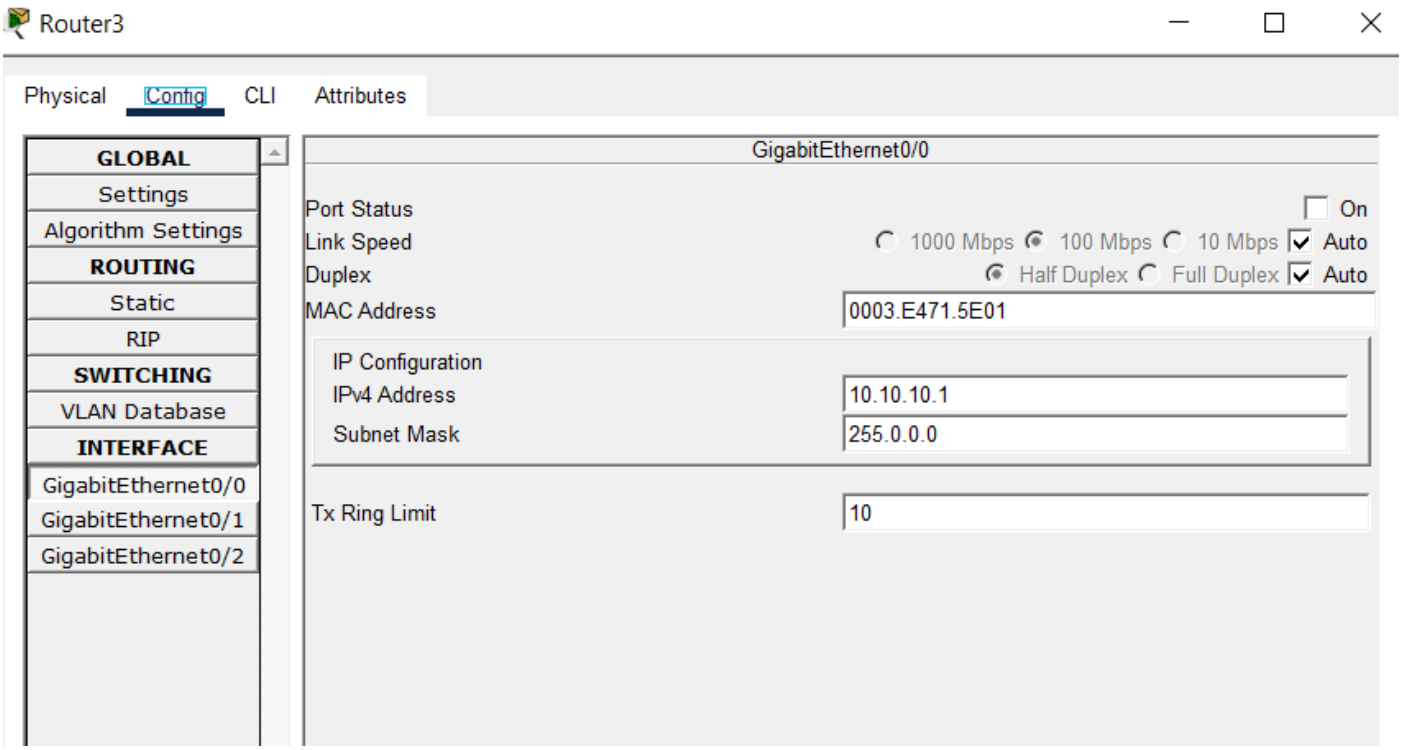
Task 9

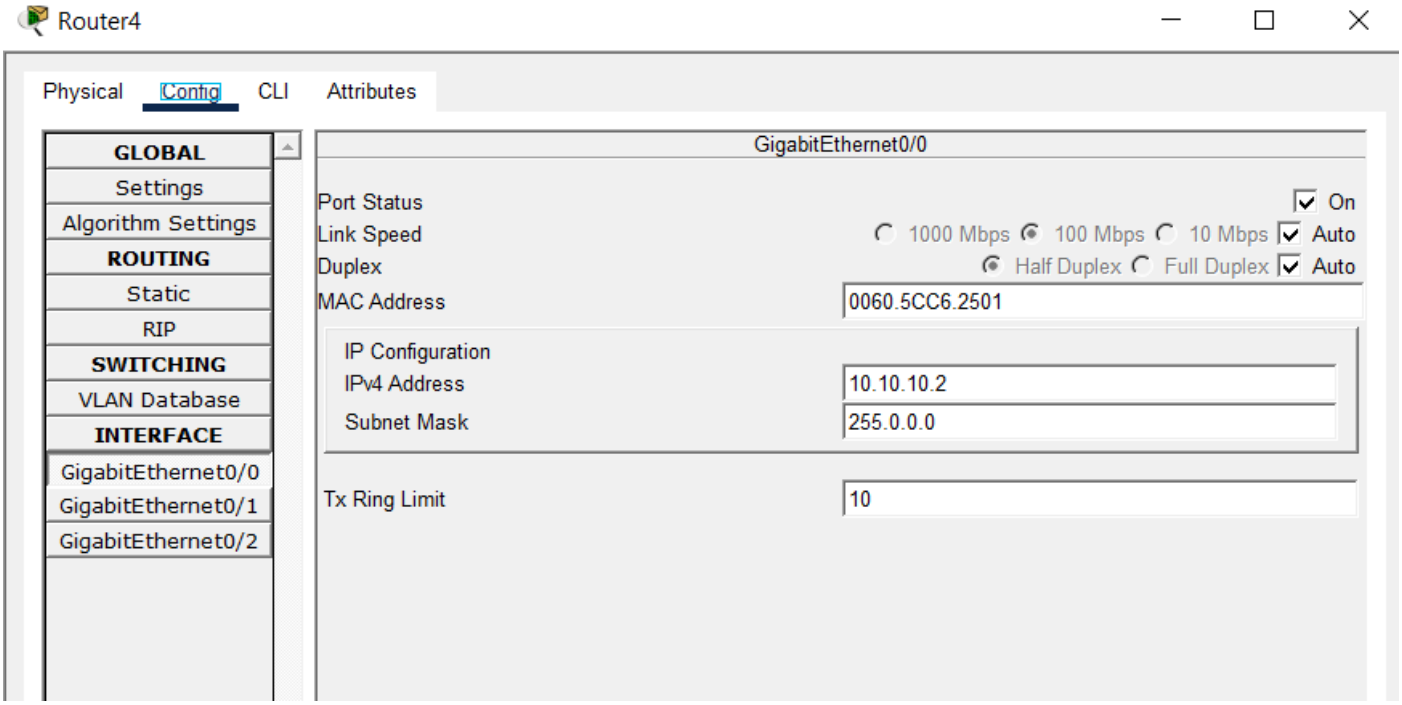
WEEK 3: Life of a Packet

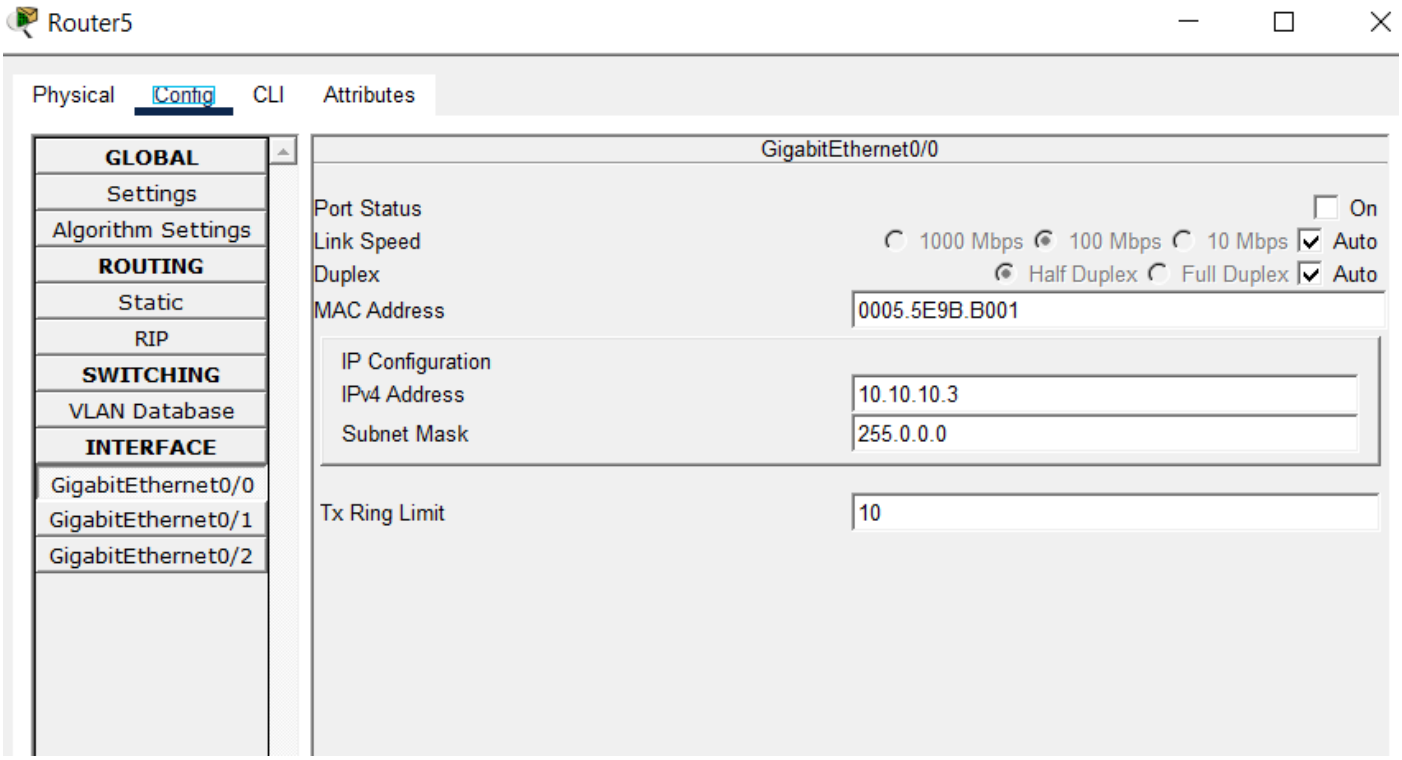


Tasks: 16

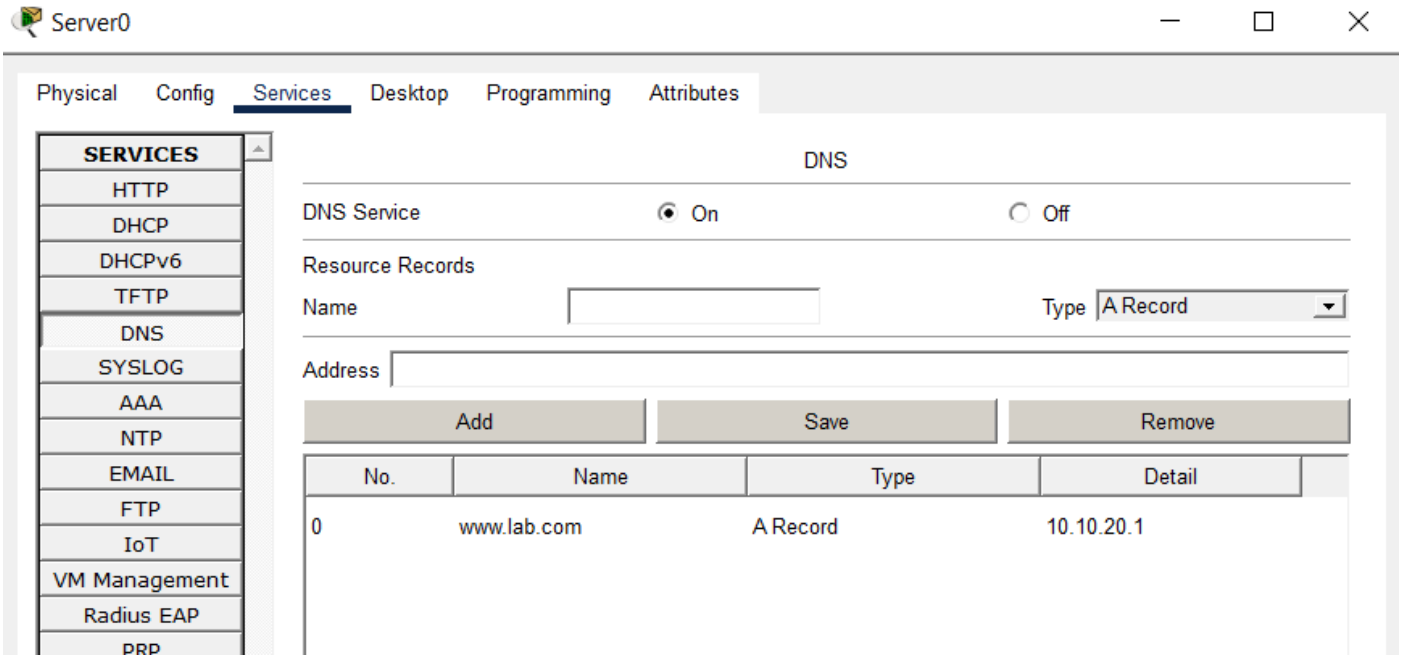
DNS Configuration




Router4
Physical
Config
CLI
Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
SWITCHING
VLAN Database
INTERFACE
GigabitEthernet0/0
GigabitEthernet0/1
GigabitEthernet0/2
GigabitEthernet0/0
Port Status
On
Link Speed
1000 Mbps
100 Mbps
10 Mbps
Auto
Duplex
Half Duplex
Full Duplex
Auto
MAC Address
0060.5CC6.2501
IP Configuration
IPv4 Address
10.10.10.2
Subnet Mask
255.0.0.0
Tx Ring Limit
10

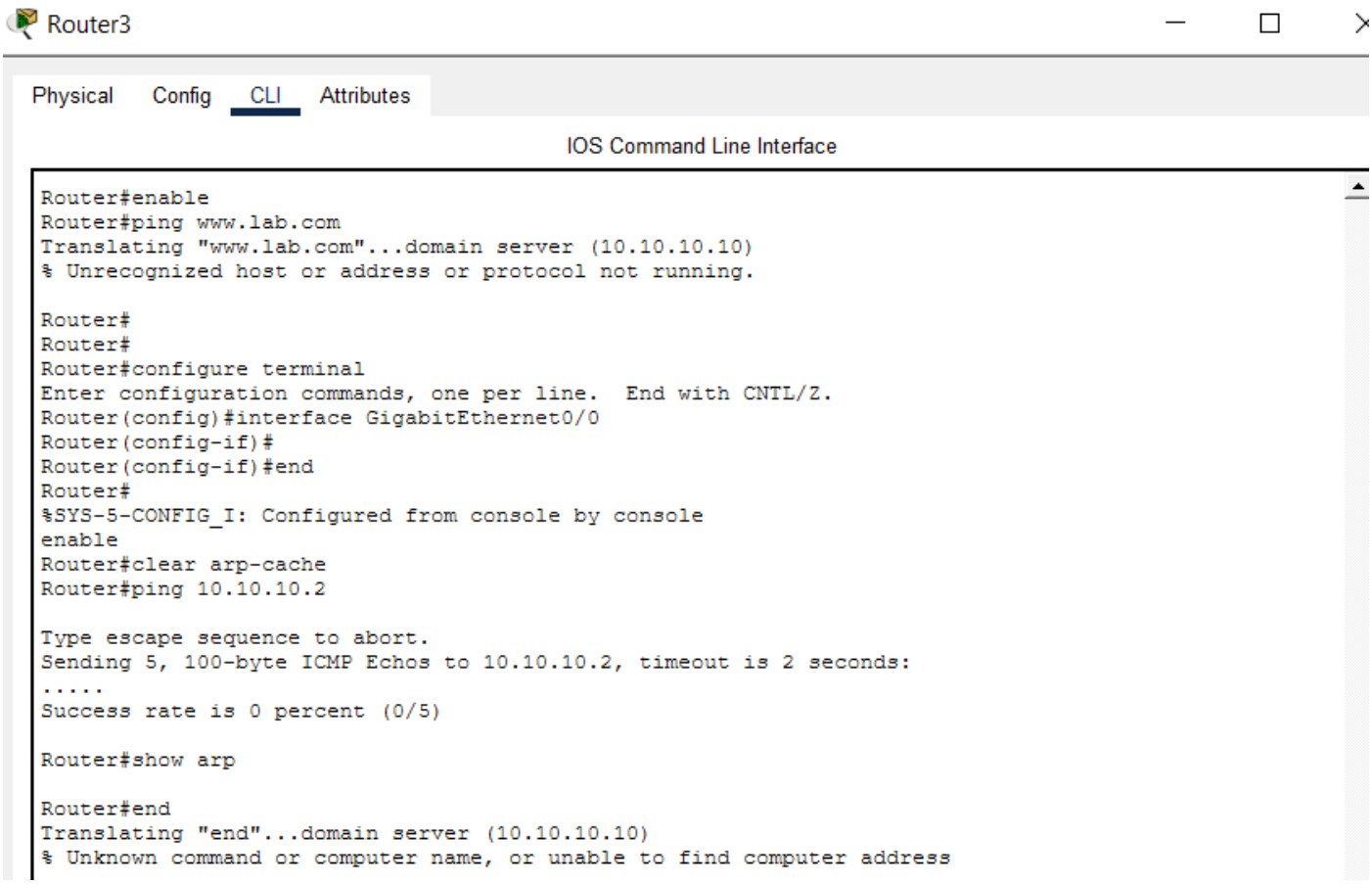

Router5
Physical
Config
CLI
Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
SWITCHING
VLAN Database
INTERFACE
GigabitEthernet0/0
GigabitEthernet0/1
GigabitEthernet0/2
GigabitEthernet0/0
Port Status
On
Link Speed
1000 Mbps
100 Mbps
10 Mbps
Auto
Duplex
Half Duplex
Full Duplex
Auto
MAC Address
0005.5E9B.B001
IP Configuration
IPv4 Address
10.10.10.3
Subnet Mask
255.0.0.0
Tx Ring Limit
10

Task 17



ARP Cache

Task 18 and 19



```
Router>enable
Router#clear arp-cache
Router#ping 10.10.20.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.20.2, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)

Router#show arp

Router#show arp

Router#enable
Router#show arp

Router#
Router#
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface GigabitEthernet0/0
Router(config-if)#
```

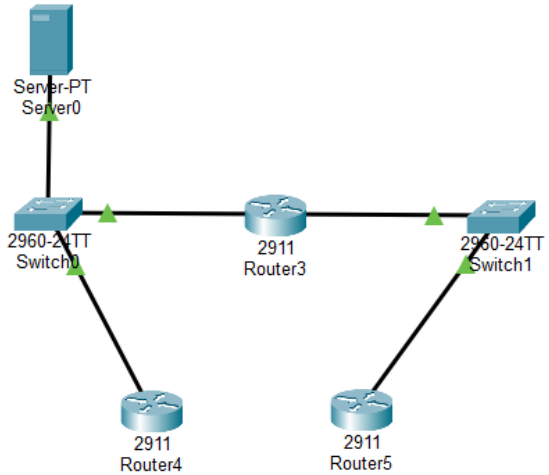
Question 20

- An ARP entry provides the MAC address associated with a particular IP address on a local network. It also identifies the interface through which the device can be reached.
- A routing table entry provides information about the destination network, the next-hop IP address and the outgoing interface. However, it does not normally provide the actual MAC address needed to deliver an Ethernet frame.

Question 21

- A router needs to perform an ARP lookup because knowing the correct route only tells the router where to forward the packet. It does not provide the destination MAC address required to transmit the Ethernet frame.
- ARP allows the router to discover the MAC address of the next-hop device. The router then places that MAC address in the destination field of the Ethernet frame before forwarding the packet.

WEEK 4: : The Cisco Troubleshooting Methodology



Task 22

```
Router>enable
Router#ping 10.10.10.10

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.10, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)

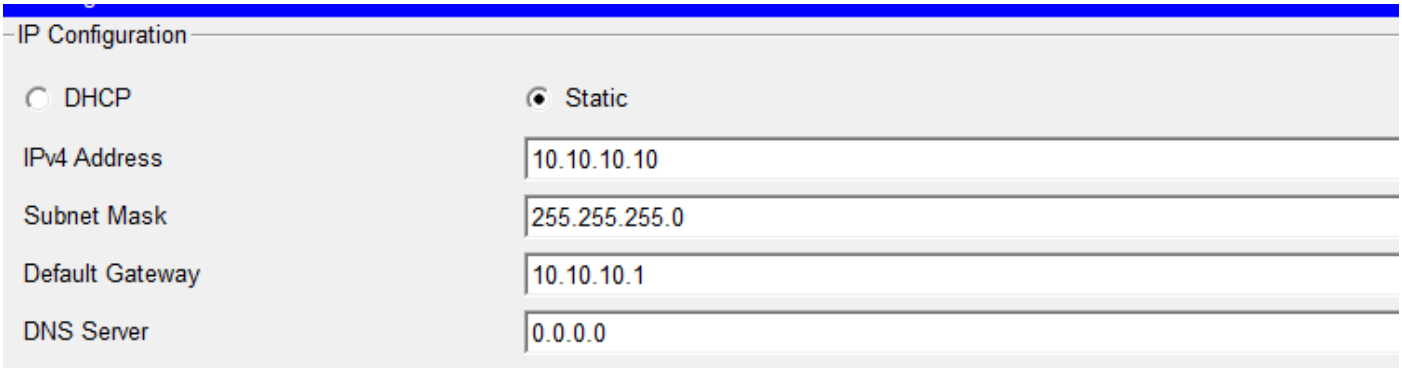
Router#
```

Task 23

```
Router#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     10.10.10.1      YES manual down                  down
GigabitEthernet0/1     unassigned      YES unset  administratively down down
GigabitEthernet0/2     unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

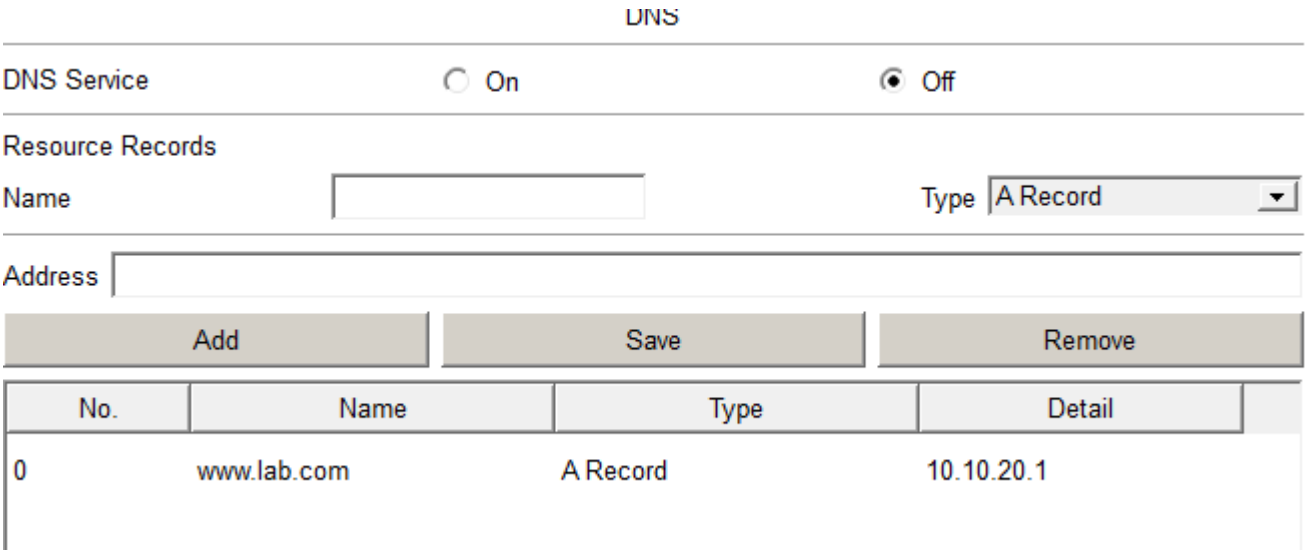
Gateway of last resort is not set
```

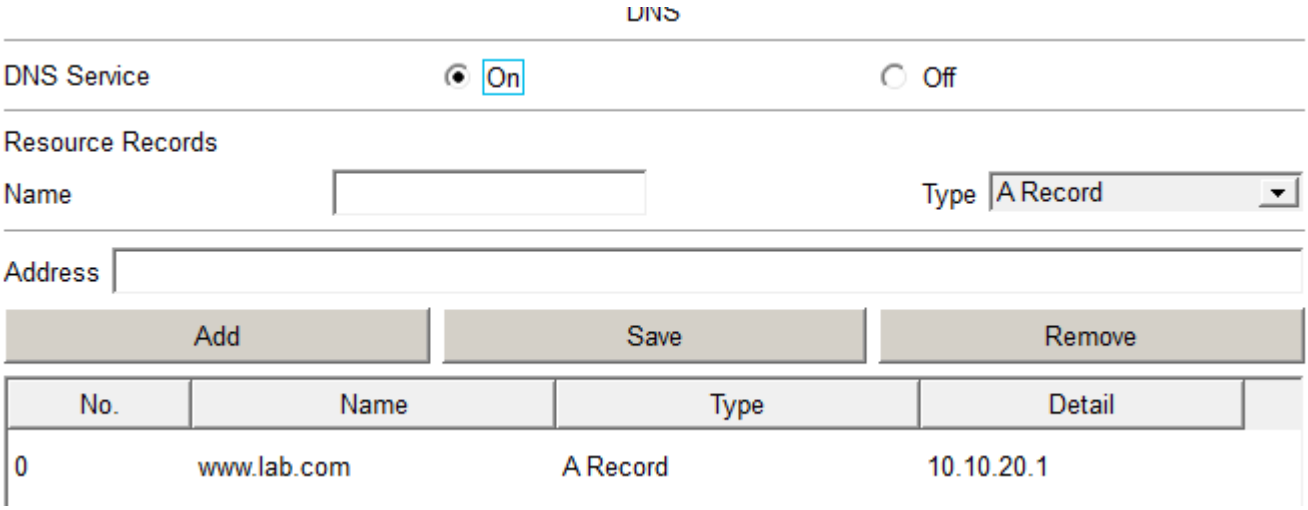
Task 24



Task 25

```
enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#ip name-server 10.10.10.10
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
```





Task 26

```
Router#ping 10.10.10.10

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.10, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)

Router#ping www.lab.com
Translating "www.lab.com"...domain server (10.10.10.10)
```

Task 27

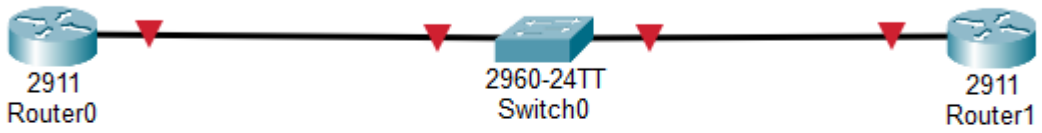
Reflection

During troubleshooting, I checked the physical connections and confirmed that the cables ad the interface status was working correctly. I then examined the network configuration and discovered that the address was incorrectly configured.

After correcting the IP address, I tested the connection again using the ping command, unfortunately it was not successful, but the DNS server had been restored.

This demonstrated the importance of following the OSI troubleshooting methodology systematically rather than making random configuration changes.

WEEK 5: Cisco Router and Switch Basics



Task 28

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname
% Incomplete command.
Router(config)#hostname R0
R0(config)#
```

Task 29

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R1
R1(config)#
```

Task 30

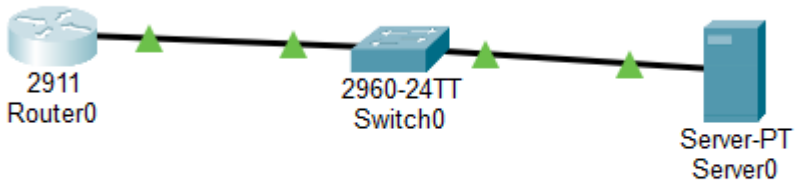
```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#hostname SW0
SW0(config)#
```

Task 31

```
R1#
R1#
R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#interface GigabitEthernet0/0
R1(config-if)#ip address 10.10.10.1 255.0.0.0
R1(config-if)#ip address 10.10.10.1 255.255.255.0
R1(config-if)#no shutdown

R1(config-if)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0, changed state to down
```

WEEK 6: Cisco Device Management



Task 48:

```
R0(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
exit
R0(config)#enable secret Cisco123
R0(config)#line console 0
R0(config-line)#password Console123
R0(config-line)#login
R0(config-line)#exit
R0(config)#end
R0#
%SYS-5-CONFIG_I: Configured from console by console
write memory
Building configuration...
[OK]
R0#ping 10.10.10.10

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.10, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/0/1 ms

R0#enable
R0#copy running-config tftp
Address or name of remote host []? 10.10.10.10
Destination filename [R0-confg]? R0-backup.cfg

Writing running-config...!!
[OK - 778 bytes]

778 bytes copied in 0.048 secs (16208 bytes/sec)
R0#show version
Cisco IOS Software, C2900 Software (C2900-UNIVERSALK9-M), Version 15.1(4)M4, RELEASE SOFTWARE (fc2)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2012 by Cisco Systems, Inc.
Compiled Thurs 5-Jan-12 15:41 by pt_team

ROM: System Bootstrap, Version 15.1(4)M4, RELEASE SOFTWARE (fc1)
cisco2911 uptime is 32 minutes, 22 seconds
System returned to ROM by power-on
System image file is "flash0:c2900-universalk9-mz.SPA.151-1.M4.bin"
Last reload type: Normal Reload
```

Task 49

```
R0#enable
R0#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R0(config)#enable secret OldPassword123
R0(config)#line console
% Incomplete command.
R0(config)#line console 0
R0(config-line)#password OldConsole123
R0(config-line)#login
R0(config-line)#exit
R0(config)#end
R0#
%SYS-5-CONFIG_I: Configured from console by console
write memory
Building configuration...
[OK]
R0#
```

```
R0(config-if)#enable
               ^
% Invalid input detected at '^' marker.

R0(config-if)#end
R0#
%SYS-5-CONFIG_I: Configured from console by console
enable
R0#show version
Cisco IOS Software, C2900 Software (C2900-UNIVERSALK9-M), Version 15.1(4)M4, RELEASE SOFTWARE (fc2)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2012 by Cisco Systems, Inc.
Compiled Thurs 5-Jan-12 15:41 by pt_team

ROM: System Bootstrap, Version 15.1(4)M4, RELEASE SOFTWARE (fc1)
cisco2911 uptime is 2 minutes, 31 seconds
System returned to ROM by power-on
System image file is "flash0:c2900-universalk9-mz.SPA.151-1.M4.bin"
Last reload type: Normal Reload

This product contains cryptographic features and is subject to United
States and local country laws governing import, export, transfer and
use. Delivery of Cisco cryptographic products does not imply
third-party authority to import, export, distribute or use encryption.
Importers, exporters, distributors and users are responsible for
compliance with U.S. and local country laws. By using this product you
agree to comply with applicable laws and regulations. If you are unable
to comply with U.S. and local laws, return this product immediately.

A summary of U.S. laws governing Cisco cryptographic products may be found at:
http://www.cisco.com/wwl/export/crypto/tool/stqrg.html
```

Task 50

```
R0#enable
R0#write erase
Erasing the nvram filesystem will remove all configuration files! Continue? [confirm]
[OK]
Erase of nvram: complete
%SYS-7-NV_BLOCK_INIT: Initialized the geometry of nvram
R0#System Bootstrap, Version 15.1(4)M4, RELEASE SOFTWARE (fc1)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 2010 by cisco Systems, Inc.
Total memory size = 512 MB - On-board = 512 MB, DIMM0 = 0 MB
CISCO2911/K9 platform with 524288 Kbytes of main memory
Main memory is configured to 72/-1(On-board/DIMM0) bit mode with ECC disabled

Readonly ROMMON initialized

program load complete, entry point: 0x80803000, size: 0x1b340
program load complete, entry point: 0x80803000, size: 0x1b340

IOS Image Load Test
___________________
Digitally Signed Release Software
program load complete, entry point: 0x81000000, size: 0x3bcd3d8
Self decompressing the image :
############################################################################ [OK]
Smart Init is enabled
smart init is sizing iomem
                  TYPE      MEMORY_REQ
     Onboard devices &
```

```
A summary of U.S. laws governing Cisco cryptographic products may be found at:
http://www.cisco.com/wwl/export/crypto/tool/stqrg.html

If you require further assistance please contact us by sending email to
export@cisco.com.

Cisco CISCO2911/K9 (revision 1.0) with 491520K/32768K bytes of memory.
Processor board ID FTX152400KS
3 Gigabit Ethernet interfaces
DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)


         --- System Configuration Dialog ---

Would you like to enter the initial configuration dialog? [yes/no]: no


Press RETURN to get started!
```

```
Router>enable
Router#show startup-config
startup-config is not present
Router#show running-config
Building configuration...

Current configuration : 696 bytes
!
version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname Router
!
!
!
!
!
!
!
!
no ip cef
no ipv6 cef
!
!
```

Task 51

```
!

Router#enable
Router#copy tftp: startup-config
Address or name of remote host []? 10.10.10.10
Source filename []? R0-backup.cfg
Destination filename [startup-config]?

Accessing tftp://10.10.10.10/R0-backup.cfg........
%Error opening tftp://10.10.10.10/R0-backup.cfg (Timed out)
Router#
```

```
Router#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     unassigned      YES unset  administratively down down
GigabitEthernet0/1     unassigned      YES unset  administratively down down
GigabitEthernet0/2     unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
Router#dir flash:
Directory of flash0:/

    3  -rw-    33591768           <no date>  c2900-universalk9-mz.SPA.151-4.M4.bin
    2  -rw-       28282           <no date>  sigdef-category.xml
    1  -rw-      227537           <no date>  sigdef-default.xml

255744000 bytes total (221896413 bytes free)
Router#copy tftp: flash:
Address or name of remote host []? 10.10.10.10
Source filename []? c2900-backup.bin
Destination filename [c2900-backup.bin]? c2900-restored.bin

Accessing tftp://10.10.10.10/c2900-backup.bin........
%Error opening tftp://10.10.10.10/c2900-backup.bin (Timed out)
Router#dir flash:
Directory of flash0:/

    3  -rw-    33591768           <no date>  c2900-universalk9-mz.SPA.151-4.M4.bin
    2  -rw-       28282           <no date>  sigdef-category.xml
    1  -rw-      227537           <no date>  sigdef-default.xml

255744000 bytes total (221896413 bytes free)
Router#show version
Cisco IOS Software, C2900 Software (C2900-UNIVERSALK9-M), Version 15.1(4)M4, RELEASE SOFTWARE (fc2)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2012 by Cisco Systems, Inc.
Compiled Thurs 5-Jan-12 15:41 by pt_team

ROM: System Bootstrap, Version 15.1(4)M4, RELEASE SOFTWARE (fc1)
cisco2911 uptime is 4 minutes, 35 seconds
System returned to ROM by power-on
System image file is "flash0:c2900-universalk9-mz.SPA.151-1.M4.bin"
Last reload type: Normal Reload

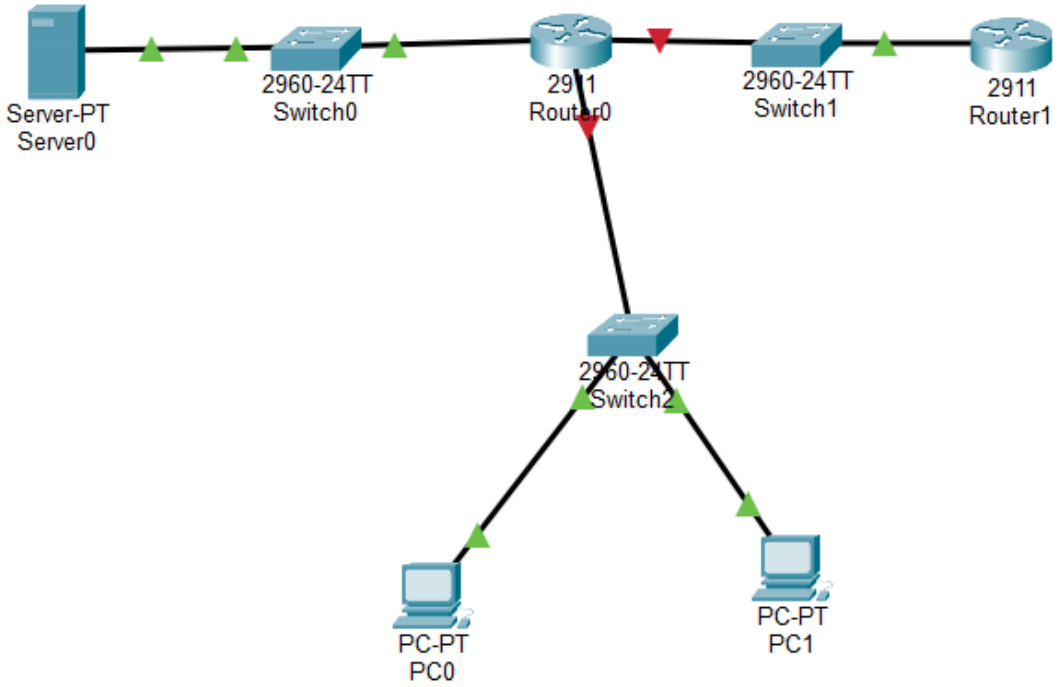
This product contains cryptographic features and is subject to United
States and local country laws governing import, export, transfer and
use. Delivery of Cisco cryptographic products does not imply
```

Conclusion

In this practical, I learned how to perform essential Cisco device management tasks using Packet Tracer. I successfully configured a TFTP server, backed up the router's running configuration and IOS image, recovered forgotten passwords, performed a factory reset, and restored the original configuration and IOS image.

This practical helped me understand the importance of configuration backups, password recovery procedures, and disaster recovery in network administration.s

WEEK 10: DHCP Configuration



Task 97